

## Probability

$$Y = g(X) \quad \Omega$$

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| $y_{13} = g(x_{13})$ | $y_{14} = g(x_{14})$ | $y_{15} = g(x_{15})$ | $y_{16} = g(x_{16})$ |
| $y_9 = g(x_9)$       | $y_{10} = g(x_{10})$ | $y_{11} = g(x_{11})$ | $y_{12} = g(x_{12})$ |
| $y_5 = g(x_5)$       | $y_6 = g(x_6)$       | $y_7 = g(x_7)$       | $y_8 = g(x_8)$       |
| $y_1 = g(x_1)$       | $y_2 = g(x_2)$       | $y_3 = g(x_3)$       | $y_4 = g(x_4)$       |

$$\Omega$$

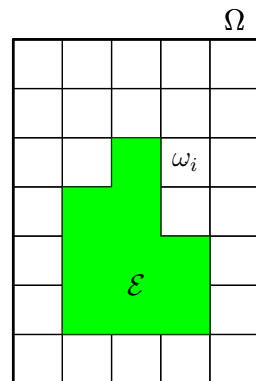
|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| $x_{31}$ | $x_{32}$ | $x_{33}$ | $x_{34}$ | $x_{35}$ |
| $x_{26}$ | $x_{27}$ | $x_{28}$ | $x_{29}$ | $x_{30}$ |
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1. Random Variables
2. Expectations
3. Calculus of Probabilities

Probability, Random Variables, Expectations

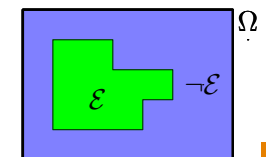
## Modelling Uncertainty

- To model a world with uncertainty we consider some set of **elementary events** or **outcomes**  $\Omega$
- For the outcome of rolling a dice  $\Omega = \{1, 2, 3, 4, 5, 6\}$
- The elementary events  $\omega_i$  are **mutually exclusive**  $\omega_i \cap \omega_j = \emptyset$  and **exhaustive**  $\bigcup_i \omega_i = \Omega$
- We consider **events**  $\mathcal{E} = \bigcup_{i \in \mathcal{I}} \omega_i$
- E.g. For a dice throw  $\mathcal{E} = \{2, 4, 6\}$



## Probabilities

- We attribute a **probability**,  $\mathbb{P}(\mathcal{E})$ , to an event,  $\mathcal{E}$ , with the requirements
  - ★  $0 \leq \mathbb{P}(\mathcal{E}) \leq 1$
  - ★  $\mathbb{P}(\mathcal{E}) + \mathbb{P}(\neg \mathcal{E}) = 1$  where  $\neg \mathcal{E} = \Omega \setminus \mathcal{E}$
- In some cases we can interpret  $\mathbb{P}(\mathcal{E})$  as the expected frequency of occurrence of a repetitive trial
- But  $\mathbb{P}(\text{Pass COMP6208 exam})$  is something you do once
- Can think of probability as an informed belief that something might happen
- When our knowledge changes the probability changes



## Random Variables

- We can define a **random variable**,  $X$ , by partition the set of outcomes  $\Omega$  and assign a numbers to each partition

- E.g. for a dice

$$X = \begin{cases} 0 & \text{if } \omega \in \{1,3,5\} \\ 1 & \text{if } \omega \in \{2,4,6\} \end{cases}$$

- $\mathbb{P}(X = x_i) = \mathbb{P}(\mathcal{E}_i)$  where  $\mathcal{E}_i$  is the event that corresponding to the partition with value  $x_i$

$\Omega$

|       |       |       |       |       |
|-------|-------|-------|-------|-------|
|       |       |       |       |       |
|       | $x_1$ |       | $x_4$ |       |
|       |       |       |       |       |
|       |       |       |       | $x_5$ |
|       |       | $x_3$ |       |       |
|       |       |       |       |       |
| $x_2$ |       |       |       |       |

## What's In A Name

- We denote random variables with capital letters,  $X$ ,  $Y$ ,  $Z$ , etc.
- The symbol denote an object that can take one of a number of different values, but which one is still to be decided by chance
- When we write  $\mathbb{P}(X)$  we can view this as short-hand for
 
$$(\mathbb{P}(X = x) \mid x \in \mathcal{X}) = (\mathbb{P}(X = x_1), \mathbb{P}(X = x_2), \dots, \mathbb{P}(X = x_n))$$
 where  $\mathcal{X}$  is the set of possible values that  $X$  can take
- We treat random variables very differently to normal numbers (scalars) when we consider taking expectations

## Function of Random Variables

- Any function,  $Y = g(X)$ , of a random variable,  $X$ , is a random variable

$Y = g(X)$   $\Omega$

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## Continuous Spaces

- If the space of elementary events is continuous (e.g. for darts  $\mathbf{x} = (x, y)$ ) then  $\mathbb{P}(\mathbf{X} = \mathbf{x}) = 0$
- But if we consider a region,  $\mathcal{R}$ , then we can assign a probability to landing in the region  $\mathbb{P}(\mathbf{X} \in \mathcal{R})$
- It is useful to work with **probability densities function** (PDF)

$$f_{\mathbf{X}}(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{\mathbb{P}(\mathbf{X} \in \mathcal{B}(\mathbf{x}, \epsilon))}{\text{Vol}(\mathcal{B}(\mathbf{x}, \epsilon))}$$



where  $\mathcal{B}(\mathbf{x}, \epsilon)$  is a ball of radius  $\epsilon$  around the point  $\mathbf{x}$  and  $\text{Vol}(\mathcal{B}(\mathbf{x}, \epsilon))$  is the volume of the ball

- If we make a change of variable the volume  $\text{Vol}(\mathcal{B}(\mathbf{x}, \epsilon))$  might change so  $f_{\mathbf{X}}(\mathbf{x})$  will change

## Change of Variables

- Consider a region  $\mathcal{R}$ —we can describe this using different coordinate systems  $x$  or  $y = g(x)$ ■

- But

$$\mathbb{P}(X \in \mathcal{R}) = \int_{\mathcal{R}} f_X(x) dx = \mathbb{P}(Y \in \mathcal{R}) = \int_{\mathcal{R}} f_Y(y) dy$$

- As this is true for any region  $\mathcal{R}$ :  $f_X(x)|dx| = f_Y(y)|dy|$ ■

- Or

$$f_X(x) = f_Y(y) \left| \frac{dy}{dx} \right| = f_Y(g(x)) |g'(x)|$$

- A probability density measured in units of probability per cm is different to that measured in units of probability per inch■

## Jacobian

- In high dimension if we make a change of variables  $x \rightarrow y(x)$  (which can be seen as a change of random variables  $X \rightarrow Y(X)$ )■

- Then

$$f_X(x) = f_Y(y) |\det(\mathbf{J})|$$

where  $\mathbf{J}$  is the Jacobian matrix

$$\mathbf{J} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \dots & \frac{\partial y_1}{\partial x_n} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \dots & \frac{\partial y_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_n}{\partial x_1} & \frac{\partial y_n}{\partial x_2} & \dots & \frac{\partial y_n}{\partial x_n} \end{pmatrix}$$

- Ensures integrals over volumes are the same■

## Meaning of Probability Densities

- Probability densities are not probabilities■
- They are positive, but don't need to be less than 1■
- Note that

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(x \leq X < x + \delta x)}{\delta x}$$

- We can think of  $f_X(x)\delta x$  as  $\mathbb{P}(x \leq X < x + \delta x)$ ■
- Note that  $f_X(x)\delta x \leq 1$ ■

## Cumulative Distribution Functions

- We can define the **cumulative distribution function** (CDF)

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} \sum_{i: x_i \leq x} \mathbb{P}(X = x_i) \\ \int_{-\infty}^x f_X(y) dy \end{cases}$$

- This is a function that goes from 0 to 1 as  $x$  goes from  $-\infty$  to  $\infty$ ■
- We note that for continuous random variables

$$f_X(x) = \frac{dF_X(x)}{dx}$$

## Outline

1. Random Variables
2. **Expectations**
3. Calculus of Probabilities

$\Omega$

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## Expectation

- We can define the expectation of  $Y = g(X)$  as

$$\mathbb{E}_X[g(X)] = \begin{cases} \sum_{x \in \mathcal{X}} g(x) \mathbb{P}(X = x) \\ \int g(x) f_X(x) dx \end{cases}$$

- The expectation of a constant  $c$  is

$$\mathbb{E}_X[c] = \begin{cases} \sum_{x \in \mathcal{X}} c \mathbb{P}(X = x) = c \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) = c \\ \int c f_X(x) dx = c \int f_X(x) dx = c \end{cases}$$

- Note  $\mathbb{E}_X[\mathbb{E}_X[g(X)]] = \mathbb{E}_X[g(X)]$

## Linearity of Expectation

- Because sums and integrals are linear operators

$$\begin{aligned} \sum_i (ax_i + by_i) &= a \left( \sum_i x_i \right) + b \left( \sum_i y_i \right) \\ \int (af(x) + bg(x)) dx &= a \left( \int f(x) dx \right) + b \left( \int g(x) dx \right) \end{aligned}$$

then expectations are linear

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

- Beware usually  $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$  (unless  $X$  and  $Y$  are independent)

## Indicator Functions

- An indicator function has the property

$$\llbracket predicate \rrbracket = \begin{cases} 1 & \text{if } predicate \text{ is True} \\ 0 & \text{if } predicate \text{ is False} \end{cases}$$

(sometimes written  $I_A(x)$  where  $A(x)$  is the predicate)

- We can obtain probabilities from expectations

$$\mathbb{P}(predicate) = \mathbb{E}[\llbracket predicate \rrbracket]$$

- E.g. The CDF is given by

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{E}[\llbracket X \leq x \rrbracket]$$

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2. Expectations
3. **Calculus of Probabilities**

$\Omega$

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- Often we want to model complex processes where we have multiple random variables

- We can define the joint probability

$$p_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y)$$

i.e. the probability of the event where both  $X = x$  and  $Y = y$

- Clearly  $\mathbb{P}(X,Y) = \mathbb{P}(Y,X)$

## Marginalisation

- Probabilities are extremely easy to manipulate (although lots of people struggle)
- One of the most useful properties is known as **marginalisation**

$$\mathbb{P}(X) = \sum_{y \in \mathcal{Y}} \mathbb{P}(X, Y = y)$$

where  $\mathcal{Y}$  is the set of values that the random variable  $Y$  takes

- Note that when we write  $\mathbb{P}(X)$  we are saying this is true for all values that  $X$  can take
- Although obvious and easy this is extremely useful

## Conditional Probability

- We can also define the probability of an event  $X$  given that  $Y = y$  has occurred

$$\mathbb{P}(X | Y = y) = \frac{\mathbb{P}(X, Y = y)}{\mathbb{P}(Y = y)}$$

- In constructing a model it is often much easier to specify conditional probabilities (because you know something) rather than joint probabilities
- When manipulating probabilities it is often easier to work with joint probabilities because we can simplify them by marginalising out random variables we are not interested in

## Basic Calculus

- To obtain the joint probability we can use

$$\mathbb{P}(X,Y) = \mathbb{P}(X|Y)\mathbb{P}(Y) = \mathbb{P}(Y|X)\mathbb{P}(X)$$

- This generalises to more random variables

$$\mathbb{P}(X,Y,Z) = \mathbb{P}(X,Y|Z)\mathbb{P}(Z) = \mathbb{P}(X|Y,Z)\mathbb{P}(Y|Z)\mathbb{P}(Z)$$

- We can do this in a number of different ways

$$\mathbb{P}(X,Y,Z) = \mathbb{P}(Y,Z|X)\mathbb{P}(X) = \mathbb{P}(Z|Y,X)\mathbb{P}(Y|X)\mathbb{P}(X)$$

- Note that  $\mathbb{P}(A,B | X,Y)$  means the probability of random variables  $A$  and  $B$  given that  $X$  and  $Y$  take particular values

## Beware

- Conditional probabilities,  $\mathbb{P}(X | Y)$  are probabilities for  $X$ , but not  $Y$

$$\sum_{x \in \mathcal{X}} \mathbb{P}(X = x | Y) = 1$$

$$\sum_{y \in \mathcal{Y}} \mathbb{P}(X | Y = y) \neq 1$$

(in general)

- Note that

$$\begin{aligned} \mathbb{E}_Y[\mathbb{P}(X | Y)] &= \sum_{y \in \mathcal{Y}} \mathbb{P}(Y = y) \mathbb{P}(X | Y = y) \\ &= \sum_{y \in \mathcal{Y}} \mathbb{P}(X, Y = y) = \mathbb{P}(X) \end{aligned}$$

## Causality

- Conditional probabilities does not imply causality
- We might have causal relationships

$$\mathbb{P}(\text{pass} | \text{study}) = 0.9 \quad \mathbb{P}(\text{pass} | \neg \text{study}) = 0.2$$

- But if we know  $\mathbb{P}(\text{study}) = 0.8$  then we can compute

$$\mathbb{P}(\text{pass}, \text{study}) = \mathbb{P}(\text{pass} | \text{study})\mathbb{P}(\text{study}) = 0.9 \times 0.8 = 0.72$$

$$\mathbb{P}(\text{pass}, \neg \text{study}) = \mathbb{P}(\text{pass} | \neg \text{study})\mathbb{P}(\neg \text{study}) = 0.2 \times 0.2 = 0.04$$

and

$$\begin{aligned} \mathbb{P}(\text{study} | \text{pass}) &= \frac{\mathbb{P}(\text{pass}, \text{study})}{\mathbb{P}(\text{pass})} \\ &= \frac{\mathbb{P}(\text{pass}, \text{study})}{\mathbb{P}(\text{pass}, \text{study}) + \mathbb{P}(\text{pass}, \neg \text{study})} = \frac{0.72}{0.72 + 0.04} \approx 0.947 \end{aligned}$$

## Independence

- Random variables  $X$  and  $Y$  are said to be **independent** if

$$\mathbb{P}(X,Y) = \mathbb{P}(X)\mathbb{P}(Y)$$

- Because  $\mathbb{P}(X,Y) = \mathbb{P}(X|Y)\mathbb{P}(Y)$  and  $\mathbb{P}(X,Y) = \mathbb{P}(Y|X)\mathbb{P}(X)$  independence implies

$$\mathbb{P}(X|Y) = \mathbb{P}(X) \quad \mathbb{P}(Y|X) = \mathbb{P}(Y)$$

- Probabilistic independence implies a mathematical co-incident not necessarily causal independence
- However causal independence implies probabilistic independence
- If  $X \in \{0,1\}$  represents the outcome of tossing a coin and  $Y \in \{1,2,3,4,5,6\}$  the outcome of rolling a dice then  $X$  and  $Y$  are independent

## Well Conducted Experiments

- In well conducted experiments we expect the results we obtain are independent■
- Let  $\mathcal{D} = (X_1, X_2, \dots, X_m)$  represents possible outcomes from a set of  $m$  well conducted experiments then

$$\mathbb{P}(\mathcal{D}) = \prod_{i=1}^m \mathbb{P}(X_i) \blacksquare$$

- Denoting a possible sentence I might say by  $\mathcal{S} = (W_1, W_2, \dots, W_m)$  then

$$\mathbb{P}(\mathcal{S}) \neq \prod_{i=1}^m \mathbb{P}(W_i) \blacksquare$$

otherwise it's time I retired■

## Conditional Independence

- Let  $K(d)$  be a random variable measuring the amount you know about ML on day  $d$  of your revision■
- From you revision schedule you can write down your belief

$$\mathbb{P}(K(d) \mid K(d-1), K(d-2), \dots, K(1)) \blacksquare$$

- But a very reasonable model is

$$\mathbb{P}(K(d) \mid K(d-1), K(d-2), \dots, K(1)) = \mathbb{P}(K(d) \mid K(d-1))$$

what you are going to know today will just depend on what you knew yesterday■

- We say that  $K(d)$  is **conditionally independent** on  $K(d-2)$ ,  $K(d-3)$ , etc. given  $K(d-1)$ ■

## Conclusion

- To work with probabilities you need to know
  - ★ How to go back and forward between joint probabilities and conditional probabilities
  - ★ How to marginalise out variables■
- You need to understand that for continuous outcomes, it makes sense to talk about the probability density■
- You need to know that expectations are linear operators and the expectation of a constant is the constant■
- You need to understand independence■